

Search & Recommendation Essentials

A Comprehensive Interview Reference

Volume III of the ML Interview Program

Interview Preparation Guide

August 27, 2026

Contents

I	The Search Stack	1
1	Search Systems and Query Understanding	3
1.1	The Production Search Stack	3
1.2	Head, Torso, and Tail Queries	3
1.3	Query Classification and Intent	3
1.4	Spell Correction	3
1.5	Query Segmentation and Tagging	3
1.6	Entity Resolution	3
1.7	Query Rewriting, Expansion, and Relaxation	3
1.8	Interview Questions	3
2	Lexical Retrieval and Inverted Indexes	5
2.1	Inverted Index Mechanics	5
2.2	BM25 as a System: Parameters and Tuning	5
2.3	Field Weighting and BM25F	5
2.4	The Analysis Chain	5
2.5	Top-k Evaluation: WAND and Block-Max	5
2.6	Learned Sparse Retrieval	5
2.7	Interview Questions	5
II	Vector & Neural Retrieval	7
3	Vector Retrieval Theory	9
3.1	Problem Statements: NN, Cosine, and MIPS	9
3.2	Why High Dimensions Are Hard	9
3.3	Trees and Branch-and-Bound	9
3.4	Locality-Sensitive Hashing	9
3.5	Graph Indexes: HNSW and DiskANN	9
3.6	Clustering and IVF	9
3.7	Quantization: PQ, OPQ, and Anisotropic	9
3.8	Sketching and Random Projections	9
3.9	Interview Questions	9

4	Neural Retrieval and Reranking	11
4.1	Bi-Encoders and Training Recipes	11
4.2	Late Interaction: ColBERT	11
4.3	Cross-Encoders and LLM Rerankers	11
4.4	Hard Negative Mining	11
4.5	Hybrid Retrieval and Fusion	11
4.6	Embedding Lifecycle in Production	11
4.7	Interview Questions	11
III	Ranking & Recommendations	13
5	Learning to Rank	15
5.1	Pointwise, Pairwise, Listwise	15
5.2	RankNet, LambdaRank, LambdaMART	15
5.3	Features and Training Data Construction	15
5.4	Position Bias and Unbiased LTR	15
5.5	Neural LTR and the GBDT Question	15
5.6	Interview Questions	15
6	Recommendation Systems	17
6.1	(Chapter moves here from DL Volume ch. 12 in Phase C — placeholder)	17
IV	Evaluation & Production	19
7	Search and RecSys Evaluation	21
7.1	Offline: NDCG Regimes and Judgment Lists	21
7.2	Relevance Judgment Programs	21
7.3	Online: Search-Native Metrics	21
7.4	Interleaving	21
7.5	A/B for Ranking Systems	21
7.6	Interview Questions	21
8	Production Retrieval and RAG Integration	23
8.1	The Multi-Stage Funnel: Latency and Cost Budgets	23
8.2	Index Freshness and Updates	23
8.3	Caching	23
8.4	Retrieval Quality Monitoring	23
8.5	RAG’s Retrieval Stage	23
8.6	Interview Questions	23

Part I

The Search Stack

Chapter 1

Search Systems and Query Understanding

TL;DR

Placeholder — this chapter is written in Phase C from the vector-retrieval digest and applied-search track design (see [review/search-recsys/](#) in the DL repo).

1.1 The Production Search Stack

1.2 Head, Torso, and Tail Queries

1.3 Query Classification and Intent

1.4 Spell Correction

1.5 Query Segmentation and Tagging

1.6 Entity Resolution

1.7 Query Rewriting, Expansion, and Relaxation

1.8 Interview Questions

Chapter 2

Lexical Retrieval and Inverted Indexes

TL;DR

Placeholder — this chapter is written in Phase C from the vector-retrieval digest and applied-search track design (see [review/search-recsys/](#) in the DL repo).

2.1 Inverted Index Mechanics

2.2 BM25 as a System: Parameters and Tuning

2.3 Field Weighting and BM25F

2.4 The Analysis Chain

2.5 Top-k Evaluation: WAND and Block-Max

2.6 Learned Sparse Retrieval

2.7 Interview Questions

Part II

Vector & Neural Retrieval

Chapter 3

Vector Retrieval Theory

TL;DR

Placeholder — this chapter is written in Phase C from the vector-retrieval digest and applied-search track design (see [review/search-recsys/](#) in the DL repo).

3.1 Problem Statements: NN, Cosine, and MIPS

3.2 Why High Dimensions Are Hard

3.3 Trees and Branch-and-Bound

3.4 Locality-Sensitive Hashing

3.5 Graph Indexes: HNSW and DiskANN

3.6 Clustering and IVF

3.7 Quantization: PQ, OPQ, and Anisotropic

3.8 Sketching and Random Projections

3.9 Interview Questions

Chapter 4

Neural Retrieval and Reranking

TL;DR

Placeholder — this chapter is written in Phase C from the vector-retrieval digest and applied-search track design (see [review/search-recsys/](#) in the DL repo).

4.1 Bi-Encoders and Training Recipes

4.2 Late Interaction: ColBERT

4.3 Cross-Encoders and LLM Rerankers

4.4 Hard Negative Mining

4.5 Hybrid Retrieval and Fusion

4.6 Embedding Lifecycle in Production

4.7 Interview Questions

Part III

Ranking & Recommendations

Chapter 5

Learning to Rank

TL;DR

Placeholder — this chapter is written in Phase C from the vector-retrieval digest and applied-search track design (see [review/search-recsys/](#) in the DL repo).

5.1 Pointwise, Pairwise, Listwise

5.2 RankNet, LambdaRank, LambdaMART

5.3 Features and Training Data Construction

5.4 Position Bias and Unbiased LTR

5.5 Neural LTR and the GBDT Question

5.6 Interview Questions

Chapter 6

Recommendation Systems

TL;DR

Placeholder — this chapter is written in Phase C from the vector-retrieval digest and applied-search track design (see [review/search-recsys/](#) in the DL repo).

6.1 (Chapter moves here from DL Volume ch. 12 in Phase C — placeholder)

Part IV

Evaluation & Production

Chapter 7

Search and RecSys Evaluation

TL;DR

Placeholder — this chapter is written in Phase C from the vector-retrieval digest and applied-search track design (see [review/search-recsys/](#) in the DL repo).

7.1 Offline: NDCG Regimes and Judgment Lists

7.2 Relevance Judgment Programs

7.3 Online: Search-Native Metrics

7.4 Interleaving

7.5 A/B for Ranking Systems

7.6 Interview Questions

Chapter 8

Production Retrieval and RAG Integration

TL;DR

Placeholder — this chapter is written in Phase C from the vector-retrieval digest and applied-search track design (see [review/search-recsys/](#) in the DL repo).

8.1 The Multi-Stage Funnel: Latency and Cost Budgets

8.2 Index Freshness and Updates

8.3 Caching

8.4 Retrieval Quality Monitoring

8.5 RAG's Retrieval Stage

8.6 Interview Questions
